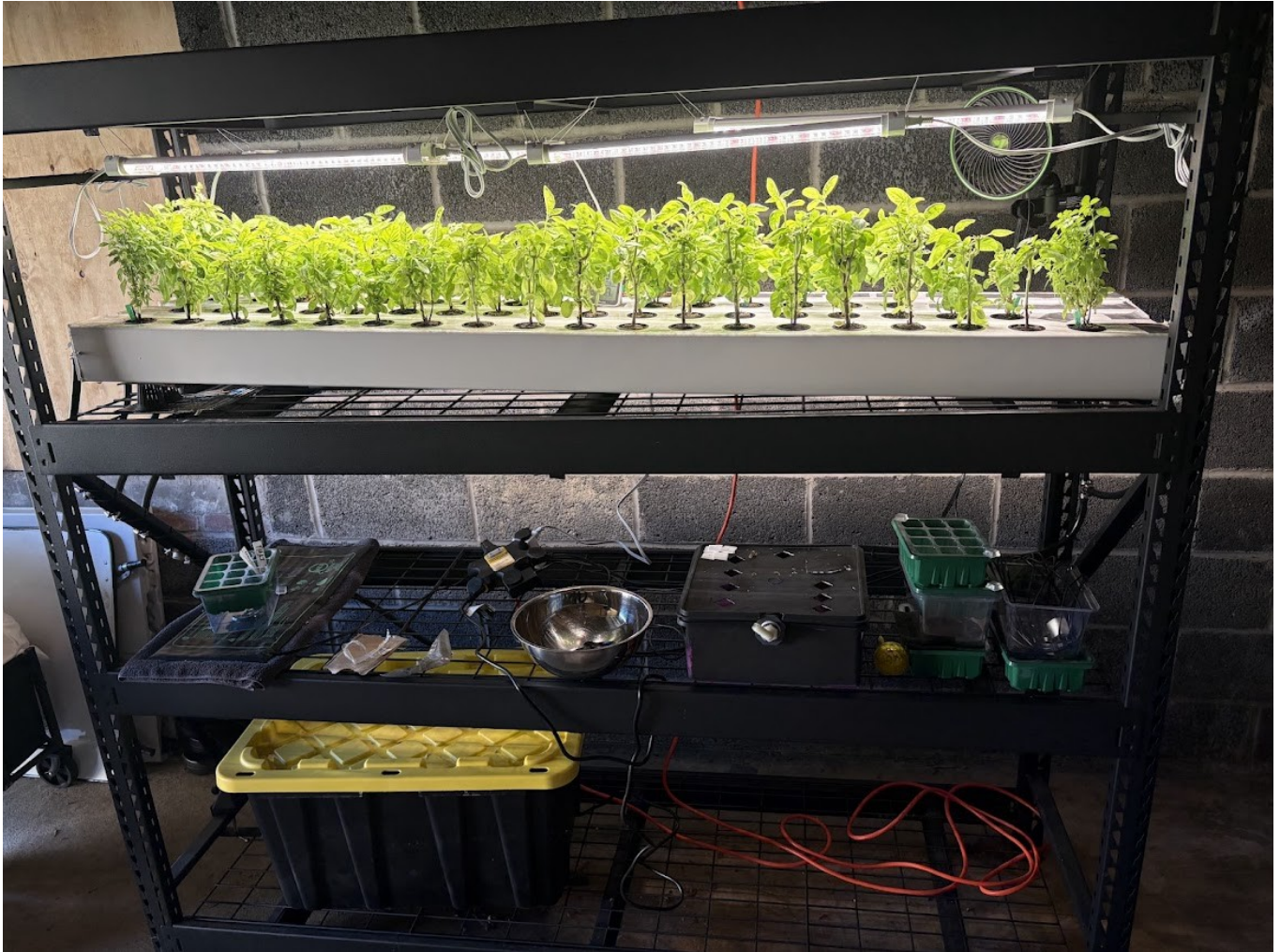


CAN A GARAGE BECOME A FARM?

Six months building, instrumenting, and operating a hydroponic basil pilot

What worked, what failed, what the economics said, and where the idea goes next



The completed 76-site pilot shelf operating in a detached one-car garage.

76

pilot NFT sites

24

recorded harvests

7.65 lb

recorded fresh biomass

~10 min/day

routine attention

Garage Basil Growth Lab | Morristown, New Jersey | March–September 2026

The short version

I was not trying to become a basil expert. I wanted to know whether an unused one-car garage could become a low-labor farm capable of producing meaningful income. Rather than answer that question only with a spreadsheet, I built the system and ran it for roughly six months.

The pilot was a four-channel, 76-site NFT shelf growing mostly Genovese basil plus Greek, Thai, Lemon, Opal, and Holy basil. I logged harvests, occupancy, reservoir events, temperature and humidity, system interventions, and a camera-based canopy record. The farm survived real garage conditions, including summer temperatures around 99°F, while mature Genovese plants continued to regrow and did not require a replacement campaign.

Bottom line

The growing platform worked. The original commodity-basil business model did not. The strongest next hypothesis is to monetize the living growing system itself—installed and managed where the food is consumed—rather than compete mainly on price per pound.

1. The original bet

The original question was bigger than “can basil grow hydroponically?” I was testing whether a smart, modular farm could convert otherwise-unused space into sustainable owner income with very little daily labor. Restaurants were the first target because they buy herbs repeatedly and care about consistency and forecastability.

Early planning assumptions were optimistic. At different points the project modeled roughly 0.04 lb of basil per site per week and much larger garage-scale outputs than the finished system ultimately demonstrated. That optimism was useful: it gave the pilot a concrete commercial hypothesis to challenge.

2. What I built



Side view of the pilot: one productive shelf, shared rack structure, reservoir below, and room intentionally left for propagation and future expansion.

Pilot geometry	1 shelf-equivalent; 4 channels × 19 sites = 76 nominal sites
Rack	77 in × 72 in × 24 in
Channels	Four 6-ft NFT channels, ~3.5 in site spacing
Reservoir	27-gal container; normally operated around 10 gal down to ~2–2.5 gal
Lighting	Four 19 W grow fixtures; 16 hr/day production schedule
Pump	Creekstone 264 GPH; continuous operation with substantial flow margin
Air movement	6 in circulation fan; effectively continuous late in the run
Instrumentation	Temperature/RH logging, Raspberry Pi, camera/canopy analysis, structured farm-state logs
Cultivars	Genovese, Greek, Thai, Lemon, Opal, Holy basil



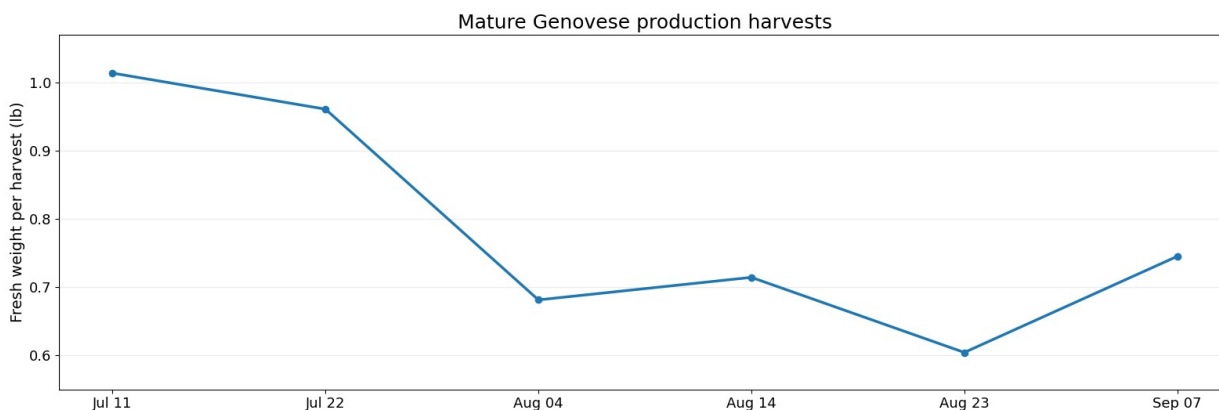
Fabrication was deliberately simple: 6-ft channels were cut, drilled for plant sites, capped, mounted, plumbed, and then iterated in-place. A repeat build is now estimated at roughly one working day, with 6–8 hours plausible after refinement.

A useful geometry correction

The final physical definition is 76 sites per shelf-equivalent and 2.5 shelf-equivalents per full rack, or 190 nominal sites. Older 99-site and 152-site projections were earlier design generations and should not be mixed with the final geometry.

3. What six months of operation showed

7.65 lb total recorded fresh biomass	~51 mature Genovese production plants	0.064 lb/day mature Genovese cohort	~40 min typical major harvest
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Repeated production harvests from essentially the same mature ~51-plant Genovese cohort.

The system became meaningfully productive in mid-June. A clean mature Genovese cohort from July 11 through September 7 produced 3.71 lb over 58 days, equivalent to about 0.064 lb/day across ~51 productive plants, or 0.00125 lb/site/day.

Day-to-day operation was much less demanding than the build and software work. Routine attention was about ten minutes per day for a walkthrough and cleanup, plus periodic reservoir service and harvests. The farm could be left alone for multi-day trips.

4. What worked better than expected

- Heat tolerance: the garage reached roughly 99°F, yet production continued and Genovese did not bolt.
- Plant persistence: mature plants were not being routinely replaced; occupancy changes were mostly deliberate additions and layout choices.
- Low operating burden: once established, the system behaved more like an appliance than a daily gardening chore.
- Hydraulic headroom: the pump had far more flow than the four-channel pilot required, leaving room for shared infrastructure at larger rack scale.
- Specialty varieties: Opal was visually striking, Lemon was compelling in flavor, Thai was excellent in specific dishes, and Holy basil became a household favorite for tea.

From rack to kitchen



The experiment produced food, not just yield data: cleaned harvests went directly into pesto and household cooking, while specialty purple cultivars made the value of variety visible in a way a pounds-per-week model could not.

Using the harvests also changed the commercial hypothesis. Genovese remained the benchmark crop, but Opal, Lemon, Thai, and Holy basil showed that the rack could create differentiated culinary value instead of competing only on commodity weight. That experience later informed the managed-rack and curated-assortment direction.

5. The real failure mode was mechanical

The late-season problem was not crop disease. The roots grew so aggressively that they formed a dense mat across the channel bottoms—roughly half an inch thick by teardown. That mat restricted drainage, raised standing water levels, created stagnant low points, and eventually pushed water high enough to expose weaknesses in channel sealing.



The late-season root system became a hydraulic object in its own right: a continuous mat filled the channel bottom, while individual plants carried dense root masses well beyond the original design assumptions.

Drain-end leaks were patched with silicone and held through the season. The deeper design lesson is more important: a commercial V2 needs explicit root-volume and drainage management, not merely better sealing.

Engineering lesson

The system did not fail because basil would not grow. It began to fail because the plants grew enough root mass to exceed the hydraulic assumptions of the channel design.

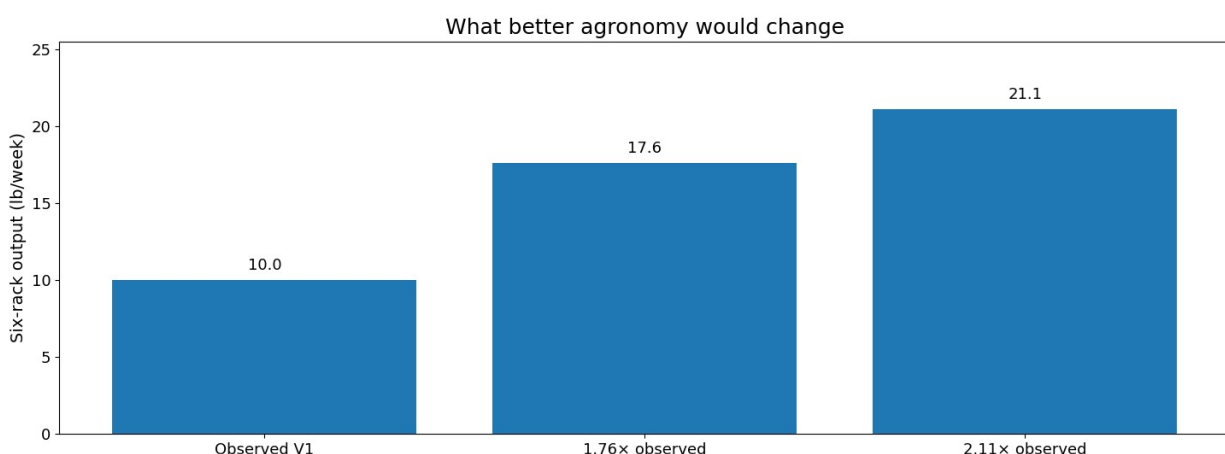
6. Was the agronomy actually bad?

No—but it was underoptimized for commercial output. The observed mature Genovese rate was ~ 0.00125 lb/site/day. Normalized across the ~ 1.19 m² shelf footprint, that is roughly 0.053 lb/m²/day from the Genovese cohort.

For context, a 2026 controlled-environment hydroponic basil study reported average temporal productivity of about 0.199 lb fresh weight/m²/day in a high-density, climate-controlled system. The systems are not directly comparable, but the benchmark is useful: V1 was biologically credible while leaving substantial room for lighting, density, nutrition, and hydraulic optimization.

Interpretation: V1 was not orders of magnitude away from serious basil production. It was a first-generation, low-input system operating well below an optimized controlled-environment benchmark. There is technical headroom—but headroom alone does not solve the business model.

7. Why the original business model did not close



Scenario output from 1,140 nominal sites; these are extrapolations, not observed garage production.

Using the observed 0.00125 lb/site/day rate, six full 190-site racks annualize to roughly 10 lb/week. That is enough to matter to individual restaurants—the project history includes one pizza restaurant reporting about 11 lb/week of basil demand—but it is not enough biomass to support a \$3,000/month owner-income target when sold like conventional basil.

Even if V2 reached roughly 1.76–2.11× the observed per-site productivity, the six-rack garage would produce only about 17.6–21.1 lb/week. Better yield helps, but the business still needs a higher-value product or a different way of charging for the system.

8. The unexpected commercial insight

The project gradually changed the question. At first I was asking how many pounds of basil a garage could sell. By the end, a more interesting possibility had emerged: the rack itself may create more value than the crop biomass.

A restaurant or household does not necessarily need the cheapest basil. It may value a managed living system that provides unusual herbs, exceptional freshness, visual appeal, a local story, and almost no work for the customer.

Model	What is sold	Main constraint
Central garage farm	Produce by weight	Yield × selling price
Private farm membership	Curated weekly harvest	Perceived household value + delivery labor
Managed living rack	Installed system + recurring service	Customer willingness to pay + serviceability

9. Direction forward: test the managed living rack

The next experiment should be commercial, not another six-month agronomy project. The working hypothesis is:

V2 commercial hypothesis

A standardized, managed hydroponic rack installed where food is consumed can create more customer value than the same rack operated as a centralized commodity farm.

Two versions are worth testing with real buyers: a chef herb rack for restaurants and a private living farm for affluent households. The rack would focus on visually and culinarily differentiated crops—potentially Genovese as a baseline plus Opal, Lemon, Thai, Holy basil, shiso, and other specialty herbs—rather than competing directly with broadline distributors on commodity basil.

The business should also be designed so routine field work does not depend on the original engineer. A general technician should be able to follow a checklist, swap standardized components, reset crops, clean the reservoir, and service the rack on a planned route. Design expertise stays in the product and procedures.

10. What I would change in V2

- Redesign drainage around mature root mass, with more cross-section and easier cleanout.
- Improve light uniformity and deliberately size PPFD/DLI instead of simply placing fixtures over occupied sites.
- Use a standardized full-rack geometry from day one rather than treating empty sites as convenient buffers.
- Keep remote monitoring simple: leak, pump/flow, temperature, and basic service alerts before advanced computer vision.
- Make pump, plumbing, reservoir, lights, and controls modular and swappable by a general technician.
- Design a repeatable propagation and crop-rotation process around the specific customer offer.
- Test willingness to pay before building a polished V2 installation.

11. What this project demonstrated

The result I am most interested in is not that I learned to grow basil. The project demonstrated an end-to-end way of working: identify a commercial hypothesis, design the physical system, build it, instrument it, operate it long enough to encounter real failure modes, collect enough data to challenge the original assumptions, and then change direction when the evidence says the business model is weaker than expected.

The takeaway

V1 proved the growing platform. V2 should test value capture and serviceability—not basic growability.



The completed pilot after six months of iteration.

Companion case study: “Garage Vision — Building a Vision Layer for a Real Farm” covers the camera, OpenCV, structured farm state, dashboard, and harvest-recovery system.

Sources and notes

- Garage Basil Growth Lab project archive and closeout records, March–September 2026.
- Pilot Costing workbook and reviewed commercial-feasibility workbook.
- Frontiers in Agronomy (2026), “Trade-offs between yield, physiological performance, and quality in basil cultivated in hydroponic versus conventional soil-based greenhouse system.” Reported hydroponic temporal productivity: ~ 0.199 lb fresh weight/m²/day.
- All garage-scale output figures are scenario extrapolations from observed mature Genovese productivity and are not observed annual production.